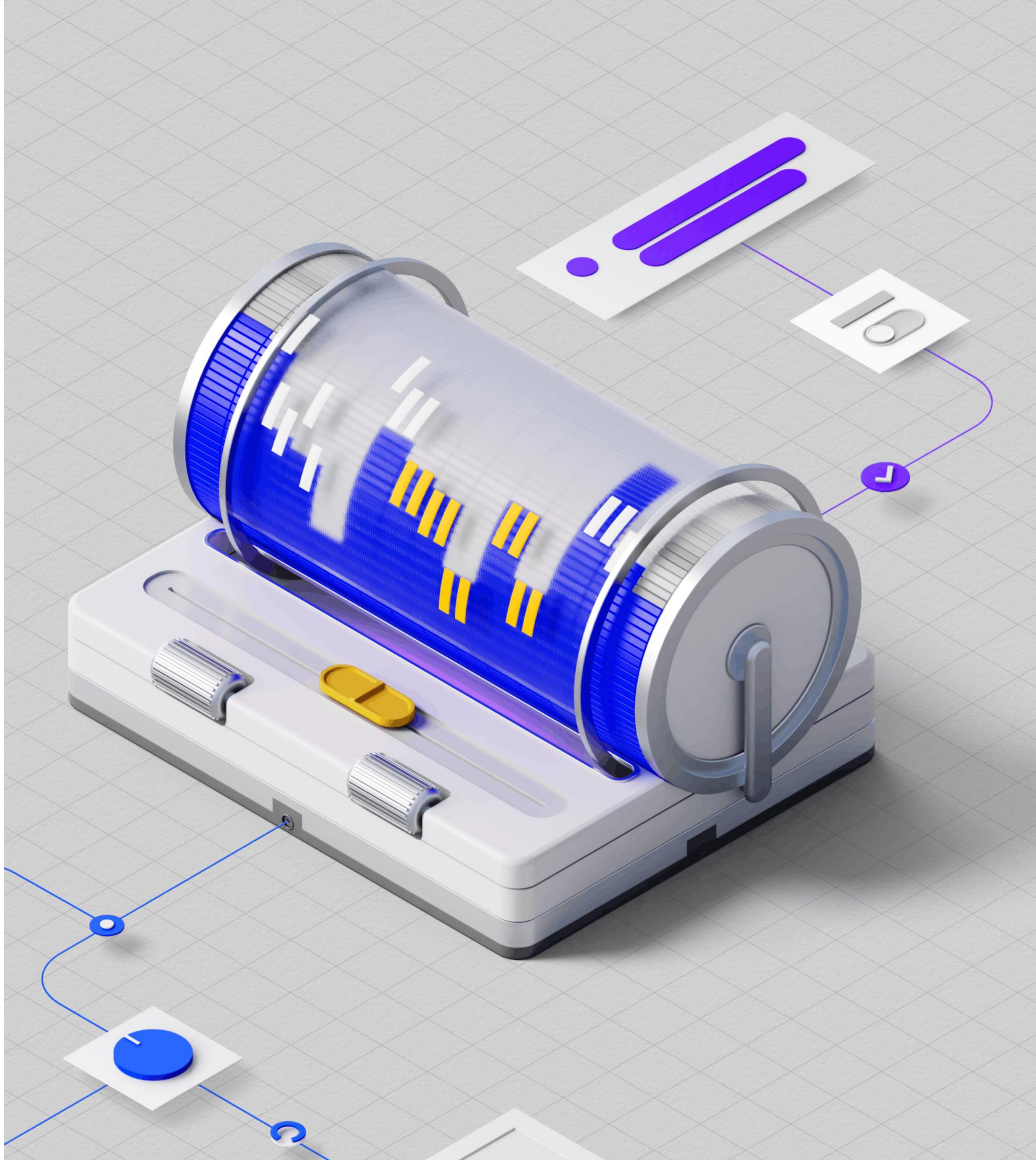
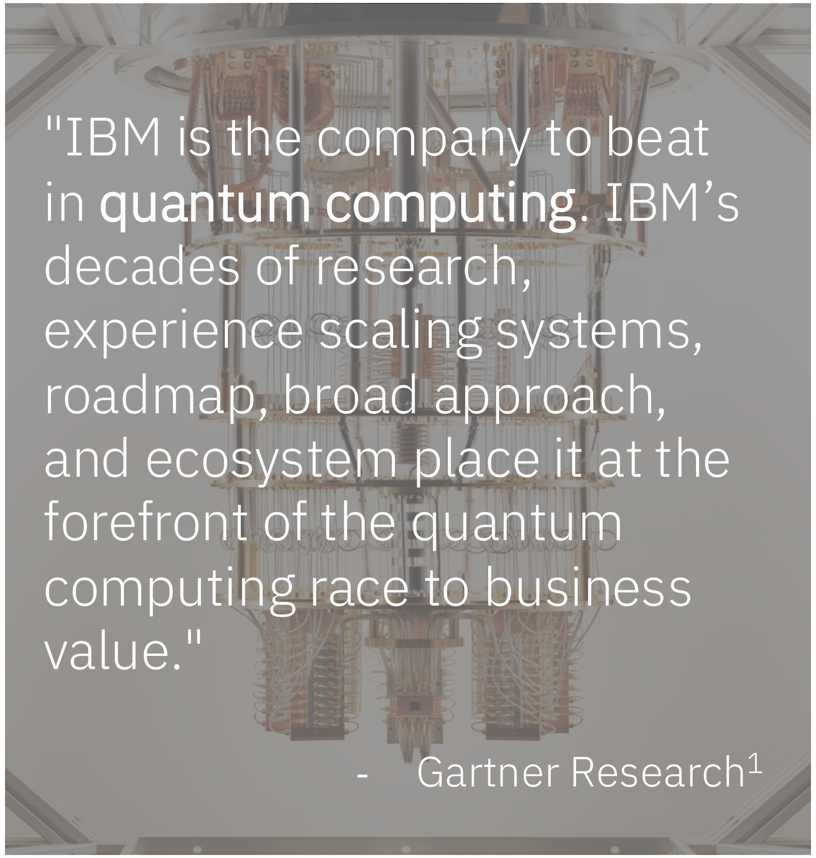
Achieving Quantum-Safe Readiness



IBM is the leader in quantum solutions

IBM leads in bringing useful quantum to the world…



"IBM is the company to beat in **quantum computing**. IBM's decades of research, experience scaling systems, roadmap, broad approach, and ecosystem place it at the forefront of the quantum computing race to business value."

- Gartner Research[1]

… and leads in making the world quantum safe

2024: IBM-Developed Algorithms Announced as NIST's First Published Post-Quantum Cryptography Standards[2]

2025: IBM's quantum-safe signature schemes advance in NIST's PQC process[3]

IBM Guardium protects[4]

4 of 5	8 of 10	7 of 10	5 of 5
Top Global Businesses	Top US Insurers	Top Global Banks	Top US Retailers

1. AI Vendor Race: IBM Is the Company to Beat in Quantum Computing, Gartner, 2025
2. IBM-Developed Algorithms Announced as NIST's First Published Post-Quantum Cryptography Standards, IBM Newsroom, 2024
3. IBM's quantum-safe signature schemes advance in NIST's PQC process, IBM Research, 2025
4. 2 Based on IBM client data and The Global 2000, Forbes, 2025

Much of today's cryptography ~~relies on hard mathematical problems~~ can be broken with a cryptographically relevant quantum computer

- ~~Public key encryption~~
- ~~Digital signatures~~
- ~~Key exchange algorithms~~

- ~~RSA~~
- ~~DSA~~
- ~~ECC~~
- ~~ECDSA~~
- ~~DH~~

Factorization

Challenge:
Find prime factors

2322787564435549164834361443028149612
9403168472741726378629043050821809225
2507358217964927292396785953285473902
2873154900644025641142681088687405784
1743673658232286043895597927098059446
654064 … 13269215
900122 … 287798
71266 … 244374
365864 … 276312
3301411 … 90502945
4640993984688031458567407248316849706
2803970194588404522298535545780490522
5263643971872454862175260511599165140
9067262683351832309929798930626982884
5288022808134366178667628733222281407
0493770251008245719155211439306593308
5829303735221526

RSA-2048

$= p \times q$

Difficulty

The most powerful computer today
would take millions of years to find the solution

Shor's quantum algorithm is anticipated to possibly break RSA in **hours** using hardware available soon

What can a cybercriminal do?

Harvest now, decrypt later

Before

Availability of cryptographically relevant quantum computers

After

Harvest confidential data to decrypt later

Decrypt lost or harvested confidential data by breaking encryption

Disrupt business with manipulation through fraudulent authentication

Manipulate digitally signed contracts and legal history by forging digital signatures

Only IBM protects sensitive data from both conventional and quantum-enabled risks

Centralizes Cryptography

No blind spots – see everything

Provides comprehensive cryptography visibility and centralized management to help organizations ensure certificates are live and encryption algorithms are up to date

→ Combats tool sprawl for cryptography management

Hybrid Approach

Assess legacy and post-quantum risk

Only IBM is built to help your organization address both conventional risks and quantum-enabled threats, guiding the transition to quantum resilient cryptography

→ Supports migration to NIST-approved quantum-safe algorithms

Current Compliance

Built to pass audits

Simplifies and expedites compliance for organizations facing existing and emerging regulations, including post-quantum cryptography standards

→ Holistic unified compliance and risk management with integrations

IBM provides a unified, comprehensive solution built to help enterprises **secure sensitive data against traditional and quantum-powered threats.**

Discover and Inventory

- Automatically discover and catalog encryption keys, certificates, algorithms, and datastores across all environments.
- Gain clear visibility with a comprehensive, consolidated dashboard of your organization's inventory of cryptographic objects

Assess and Comply

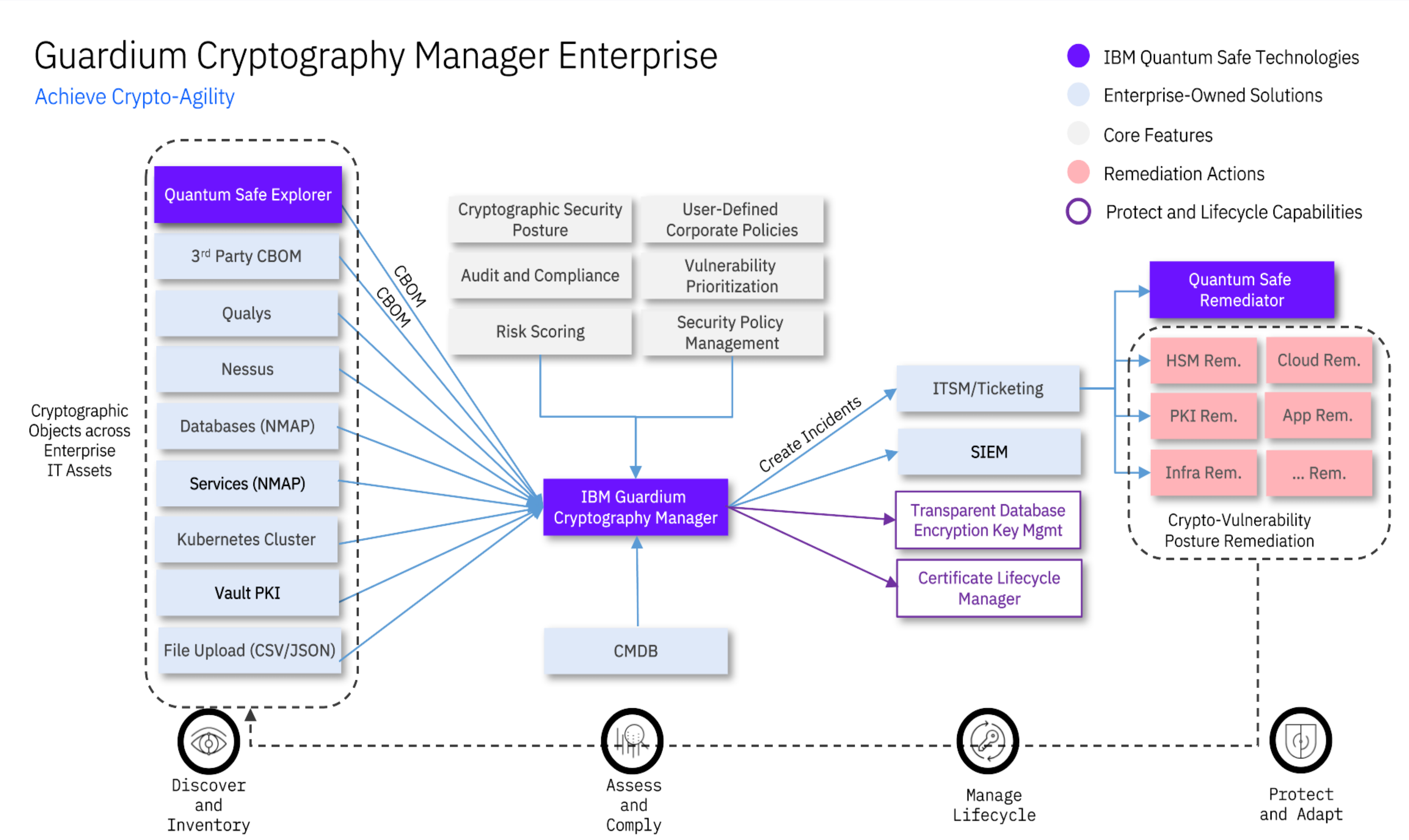
- Assess post-quantum cryptography risk, support migration to NIST-approved quantum-safe algorithms, and maintain standards alignment
- Prioritize vulnerabilities according to business risk with dependency mapping and enable automated remediation to expedite response

Manage Lifecycle

- Centrally manage key and certificate lifecycles across the environment
- Automate certificate renewals and manage revocation to streamline management to reduce operational outages

Protect and Adapt

- Simplify protection of applications and network endpoints against "harvest now, decrypt later" attacks
- Orchestrate agentless transparent database encryption key management for major databases



Quantum Safe Explorer: Automatically scans applications and infrastructure to inventory cryptographic assets and identify quantum-vulnerable algorithms.

Guardium Cryptography Manager: Prioritizes remediation based on business impact and regulatory requirements, helping teams focus on high-risk areas

Quantum Safe Remediator: Supports the transition to quantum-safe algorithms and policies, enabling crypto agility at scale

Personas to Target:

- Chief Information Security Officer (CISO)
- Chief Risk Officer (CRO)
- Chief Compliance Officer (CCO)
- Chief Privacy Officer
- Vice President of Data Security
- Security Operations Manager
- Security Architect / Platform Owner
- DevSecOps Lead

IBM Bob is an enterprise AI orchestration platform built to modernize and operate complex hybrid estates — not just generate code

What IBM Bob Enables

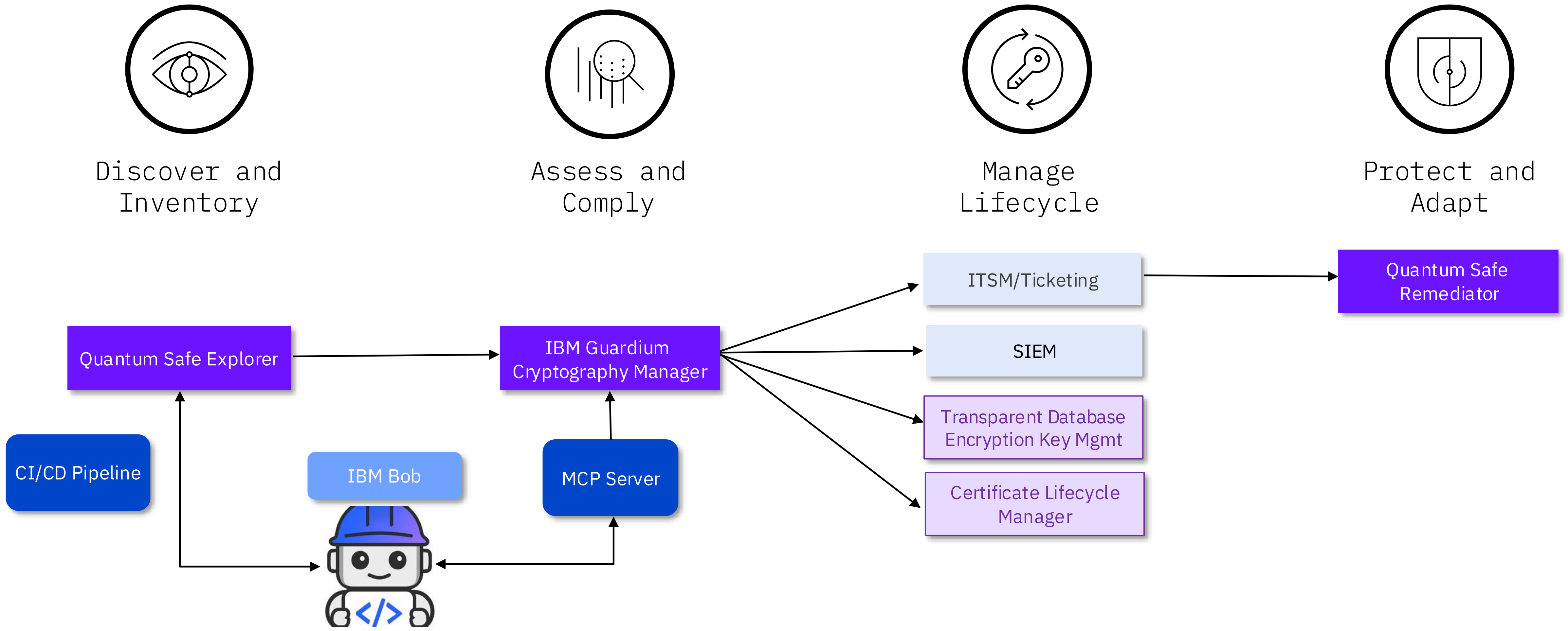
- Portfolio-level modernization
- Agentic IT operations orchestration
- Cross-platform transformation (Z, i, Java, Cloud)
- Reusable partner “Modes” as IP
- Embedded governance & compliance

What Makes It the New Standard

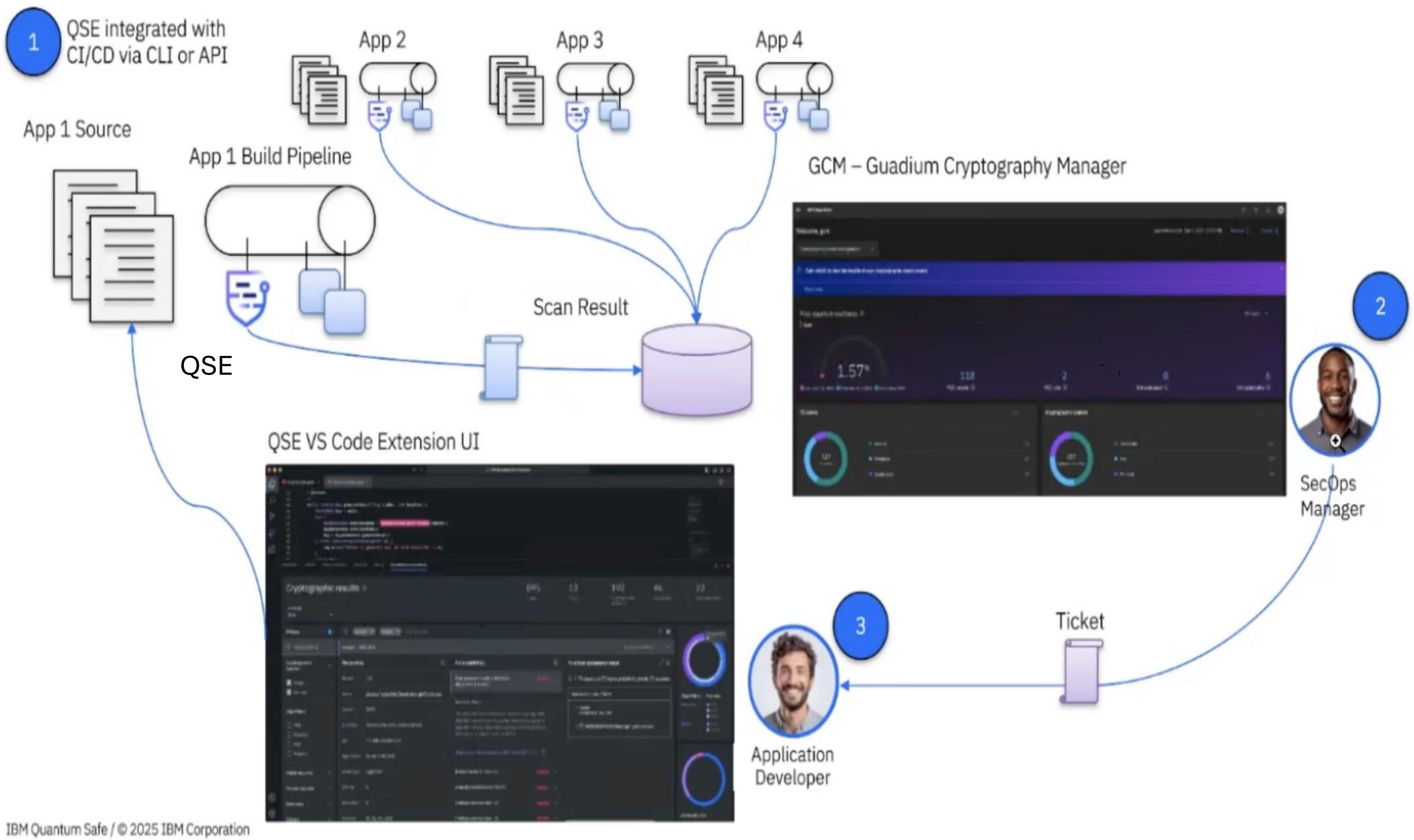
- Hybrid-first architecture
- Deep integration with enterprise tooling
- Multi-technology orchestration
- Production-grade observability
- Enterprise security by design

IBM Quantum and Bob

- Guardium Cryptography Manager Enterprise Edition
- Enterprise-Owned Solutions
- Guardium Cryptography Manager Add-on Capabilities

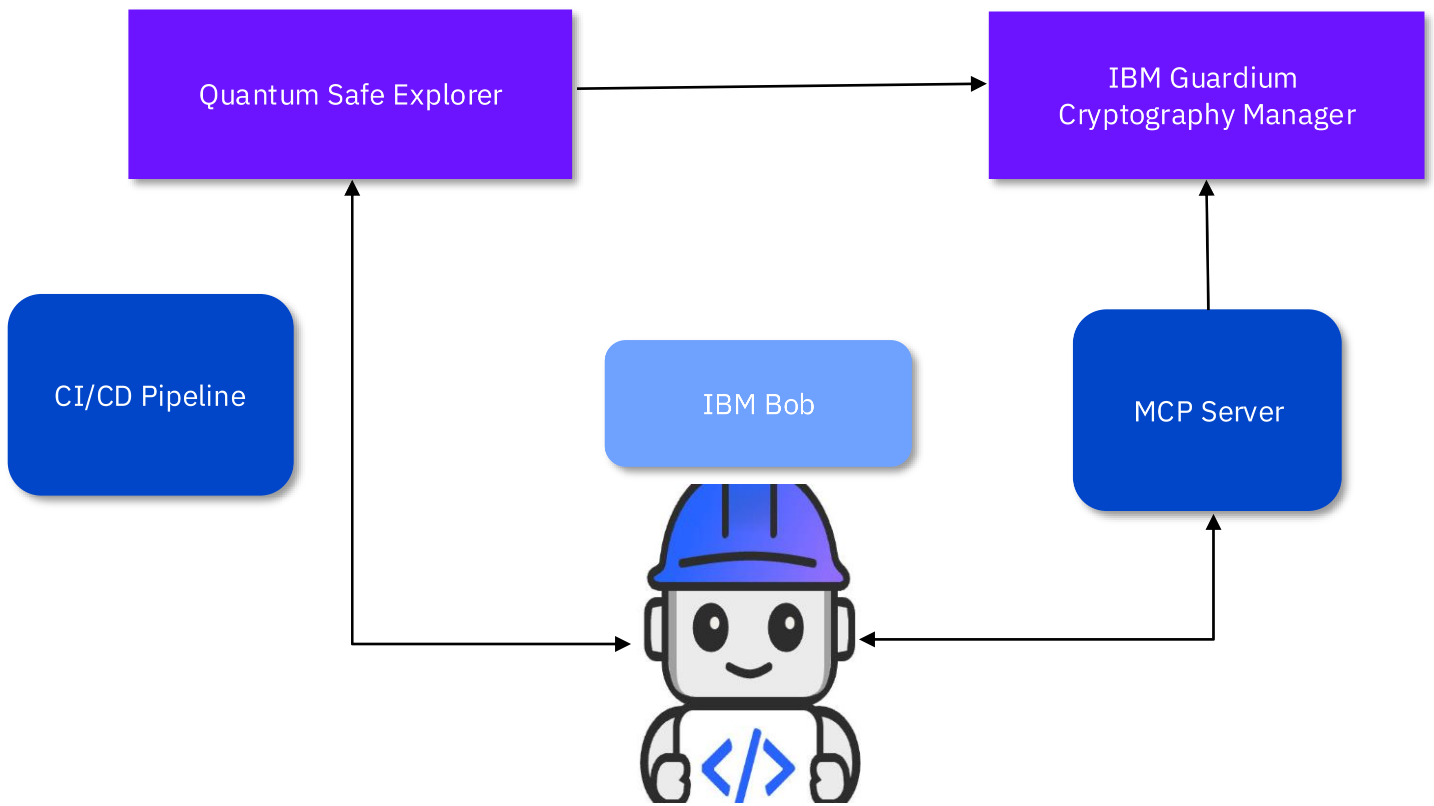


Quantum Safe Explorer (QSE) - Automated Workflow



1. QSE is integrated with CI/CD pipeline via QSE CLI app
2. QSE auto scans the code and generates Findings file containing PQC unsafe algorithms, libraries and crypt vulnerabilities
3. Finding files auto imported to GCM for enterprise view of PQC gap
4. SecOps Manager evaluates the dashboard and initiates remediation flow
5. Application Developer starts remediation with IBM Bob

Code Remediation with IBM Bob



- Application Developer initiates Remediation through IBM Bob PQC Secure Coding Mode
- This mode contains the capability for implementing PQC safe algorithms and remediate crypto vulnerabilities using latest libraries and dependencies
- IBM Bob analyses the Findings file by going through the recommendations given by GCM by connecting to its MCP server
- Bob initiates Code fixes and QSE auto scans the source code while CI/CD builds are being done
- The results are auto imported to GCM, and the dashboards show improved score on PQC readiness
- The source code changes are committed to the repository
- Any new/updated remediation measures are documented as PQC secure code best practices and IBM Bob will use them for future implementations and remediations.

Demo
Let’s see it in action!